

1. Operational Concept Description (OCD)

1.1. Project name

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1.2. Scope

The scope of this project is the development of a swarm robotics system capable of forming predefined shapes using multiple mobile robots within a controlled indoor environment. The project focuses on the coordination, positioning and movement of the robots in order to create static formations through centralized control.

The project includes the development of a centralized swarm control system, real-time robot tracking using an overhead camera, communication between the robots and a central server, and collision avoidance using the robots' onboard distance sensors. In addition, software will be developed that translates predefined shapes into robot formations. A simulation environment will also be included to validate the swarm behavior.

The system is limited to two-dimensional movement on a flat indoor surface and focuses only on predefined static formations. The swarm will operate using a centralized control architecture in which all high-level decision making is performed by a central server.

Aspects outside the scope of this project include decentralized swarm intelligence, autonomous decision making between robots, outdoor operation and dynamic or adaptive formations that change during operation.

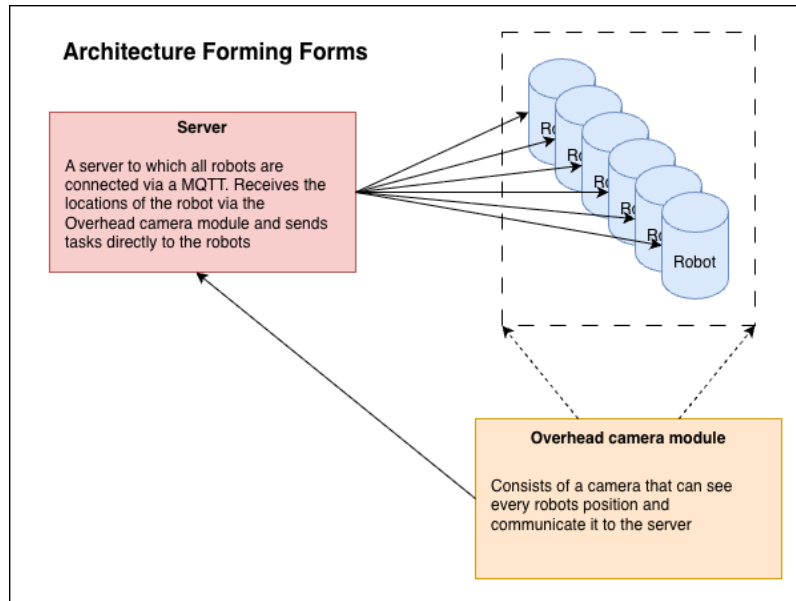
1.3. System overview

The system that is going to be built will have the goal of robots coming together to create shapes/forms.

The robots used will be the standard Chariot provided by the Hogeschool Rotterdam. The robots will be controlled by a camera, which will hang above the movement space of the robots. When a shape is inputted, the camera will use a present of this shape and compare it to the positions of the robots using object detection and give signals through a central server to steer the robots to the right place.

The thinking here will be done by the software running on the camera and the robots will react to the signals. The robots will however individually check their distance sensors to make sure they don't hit each other or other objects.

The architecture being used here is the Centralized architecture with a robot tracking system. This is the best option for the project due to there being oversight over the whole process and different robots which allows for better coordination.



Webots will be used as the simulation software, this is because this software has been used before by all the group members.

1.4. Task assignments

The project will be divided into the following subjects (responsible, supportive):

- Server (MQTT) - Hidde, Kai
- Robots (Hardware) - Daniël, Peter
- Tracking module (Object detection) - Peter, Daniël
- Path planning - Kai, Hidde
- Simulation (WeBots) – To be decided